

The sharp $2/D$ Cheeger bound in nonpositive generalized dimension

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Source and question

Milman [1] proves that a weighted manifold satisfying $\text{CDD}(0, N, D)$, with $N \in (-\infty, 0] \cup [n, \infty]$ and $D < \infty$, has

$$D_{\text{Che}, \infty}(M, g, \mu) := \inf_{0 < v < 1} \frac{\mathcal{I}(M, g, \mu)(v)}{\min(v, 1 - v)} \geq \frac{1}{D}.$$

Remark 5.20 observes that the optimal $2/D$ bound is known when $N \in [n, \infty]$, and asks whether it also holds for $N \leq 0$. The paper reduces the missing case to an explicit calculation for the one-dimensional densities t^{N-1} on $[\xi, \xi + D]$.

5.6 Linear Cheeger Constant when $D < \infty$

Finally, we have:

Theorem 5.19. *Let (M^n, g, μ) satisfy the $\text{CDD}(0, N, D)$ condition, with $N \in (-\infty, 0] \cup [n, \infty]$ and $D < \infty$. Then:*

$$D_{\text{Che}, \infty}(M, g, \mu) = \inf_{v \in (0, 1)} \frac{\mathcal{I}(M, g, \mu)(v)}{\min(v, 1 - v)} \geq \frac{1}{D}.$$

Remark 5.20. In the Euclidean setting, this was proved by Bobkov [9] Section 9], by employing the localization method. When $N \in [n, \infty]$, it is not difficult to improve this bound to $\frac{2}{D}$, which is best possible, as witnessed by considering the uniform measure on $[0, D]$; indeed, any $\text{CDD}(0, N, D)$ space for N in that range is in particular a $\text{CDD}(0, \infty, D)$ space, and using the exponential model space described in (5.15) below, this amounts to an elementary computation. We do not know if the improved bound also holds when $N \in (-\infty, 0]$, as the verification involves a tedious calculation which we did not pursue.

Answer and proof intuition

The answer is yes. For a decreasing density on an interval, the half-line Cheeger constant is the smallest reverse hazard to the right of the median. Writing $\beta = -N$, scaling the upper endpoint to one, and taking the reciprocal turns the entire calculation into maximizing

$$\phi_\beta(x) = \frac{x(1 - x^\beta)}{\beta}$$

from the median to 1; at $\beta = 0$, its continuous limit is $\phi_0(x) = x \log(1/x)$. The median is explicit. Comparing it with the unique stationary point of ϕ_β , a single tangent-line inequality shows that the maximum is at most half the interval length. Inverting gives the desired $2/D$.

Theorem 1 (Sharp negative-dimension Cheeger bound). *Let (M^n, g, μ) satisfy CDD(0, N, D), where $N \leq 0$ and $0 < D < \infty$. Then*

$$D_{\text{Che}, \infty}(M, g, \mu) \geq \frac{2}{D}.$$

The constant is optimal for every such N .

The one-dimensional calculation

We first record the exact calculation needed by the source's model theorem.

Lemma 1. *For $N \leq 0$, $\xi > 0$, and $D > 0$, let ν be the probability measure on $[\xi, \xi + D]$ with density proportional to t^{N-1} . If $\mathcal{I}_\nu^\flat$ denotes its half-line isoperimetric profile, then*

$$\inf_{0 < v < 1} \frac{\mathcal{I}_\nu^\flat(v)}{\min(v, 1-v)} \geq \frac{2}{D}.$$

Proof. Put $w(t) = t^{N-1}$, $b = \xi + D$, and $Z = \int_\xi^b w(t) dt$. Since w is decreasing, if $0 < v \leq 1/2$ and x is determined by

$$\int_x^b w(t) dt = vZ,$$

then the right half-line has the smaller of the two possible half-line boundary densities. Hence

$$\frac{\mathcal{I}_\nu^\flat(v)}{v} = \frac{w(x)}{\int_x^b w(t) dt}. \quad (1)$$

As v ranges over $(0, 1/2]$, x ranges from the median m to b . Symmetry of $\mathcal{I}_\nu^\flat$ handles $v \geq 1/2$. It follows that the reciprocal of the left-hand side in the lemma is

$$\max_{m \leq x < b} \frac{\int_x^b w(t) dt}{w(x)}. \quad (2)$$

Dilate by b , so the interval becomes $[a, 1]$, where $a = \xi/b$, and its length becomes $d = 1 - a = D/b$. The quantity in (2) scales by b . Set $\beta = -N \geq 0$, so the scaled density is $x^{-\beta-1}$. We will prove that the scaled residual quantity is at most $d/2$.

Assume first that $\beta > 0$. Direct integration gives

$$\phi_\beta(x) := \frac{\int_x^1 t^{-\beta-1} dt}{x^{-\beta-1}} = \frac{x(1 - x^\beta)}{\beta}. \quad (3)$$

The median relation is

$$m^\beta = \frac{2a^\beta}{1 + a^\beta}, \quad a = A_\beta(m), \quad A_\beta(x) := x(2 - x^\beta)^{-1/\beta}. \quad (4)$$

Moreover,

$$\phi'_\beta(x) = \frac{1 - (\beta + 1)x^\beta}{\beta}, \quad x_* = (\beta + 1)^{-1/\beta}$$

is its unique maximizer on $(0, 1)$.

We claim that

$$2\phi_\beta(x) \leq 1 - A_\beta(x) \quad (x_* \leq x \leq 1). \quad (5)$$

Indeed, let

$$F_\beta(x) = 1 - A_\beta(x) - 2\phi_\beta(x), \quad u = x^\beta.$$

Since

$$A'_\beta(x) = 2(2 - x^\beta)^{-1-1/\beta},$$

we have

$$\frac{1}{2}F'_\beta(x) = \frac{(\beta + 1)u - 1}{\beta} - (2 - u)^{-1-1/\beta}. \quad (6)$$

The function $g(u) = (2 - u)^{-1-1/\beta}$ is convex. Its tangent at $u = 1$ is

$$g(1) + g'(1)(u - 1) = \frac{(\beta + 1)u - 1}{\beta}.$$

Thus (6) is nonpositive for $u \in [1/(\beta + 1), 1]$, equivalently for $x \in [x_*, 1]$. Since $F_\beta(1) = 0$, this proves (5).

If $m \geq x_*$, then ϕ_β decreases on $[m, 1]$, and (5) together with (4) yields

$$2 \max_{[m, 1]} \phi_\beta = 2\phi_\beta(m) \leq 1 - A_\beta(m) = 1 - a = d.$$

If $m < x_*$, then $A_\beta(m) < A_\beta(x_*)$, because $A'_\beta > 0$. Using (5) at x_* gives

$$2 \max_{[m, 1]} \phi_\beta = 2\phi_\beta(x_*) \leq 1 - A_\beta(x_*) < 1 - a = d.$$

So the desired residual bound holds for every $\beta > 0$.

When $\beta = 0$, the density is x^{-1} ,

$$\phi_0(x) = x \log(1/x), \quad m = \sqrt{a}, \quad x_* = e^{-1}.$$

Set $A_0(x) = x^2$ and

$$F_0(x) = 1 - A_0(x) - 2\phi_0(x) = 1 - x^2 + 2x \log x.$$

For $x \in [x_*, 1]$,

$$F'_0(x) = 2(1 - x + \log x) \leq 0,$$

and $F_0(1) = 0$, so $F_0 \geq 0$ there. The same two cases $m \geq x_*$ and $m < x_*$, using $a = A_0(m)$, prove $\max_{[m, 1]} \phi_0 \leq (1 - a)/2$.

Undoing the dilation, (2) is at most $D/2$. Equation (1) now proves the lemma. $\square$

Passage to the manifold and sharpness

For $N \leq 0$, Case 3 of the source's model-space corollary gives, pointwise in v ,

$$\mathcal{I}(M, g, \mu)(v) \geq \inf_{\xi > 0} \mathcal{I}^b(t^{N-1}, [\xi, \xi + D])(v).$$

Dividing by $\min(v, 1 - v)$ and applying Lemma 1 to every ξ yields

$$\frac{\mathcal{I}(M, g, \mu)(v)}{\min(v, 1 - v)} \geq \frac{2}{D}$$

for every $v \in (0, 1)$. Taking the infimum proves Theorem 1.

The uniform probability measure on an interval of length D satisfies $\text{CDD}(0, N, D)$ for every $N \leq 0$: its density is constant and its generalized Ricci tensor is zero. Its isoperimetric profile equals $1/D$ on $(0, 1)$, so its linear Cheeger constant is $2/D$, attained at $v = 1/2$. The bound is therefore sharp.

Verification, novelty, and scope

The scalar formula was checked independently at 80 decimal digits on 10,000 random parameter pairs spanning twelve decades of β ; the smallest sampled value of D times the model Cheeger constant was 2.00000000000000000000000008357... This computation is only a sanity check; the proof above is exact.

On 2026-08-11, exact-phrase, exact-title, negative-dimension Cheeger/diameter, and citation searches found no later claimed resolution. The run’s lightweight indexes and the local arXiv papers citing the source also contained no matching result. This is a bounded novelty check, not an exhaustive survey of all non-arXiv literature.

The theorem answers precisely the missing $N \leq 0$ branch of Remark 5.20. Together with the already-known $N \in [n, \infty]$ branch, it improves Theorem 5.19 from $1/D$ to the sharp $2/D$ throughout that theorem's entire stated dimension range.

Human review recommendation

Review as a likely-valid full sharp solution. The highest-value checks are the half-line-profile identity (1), the tangent inequality leading to (6), and the scaling that restores D .

References

- [1] Emanuel Milman, *Beyond traditional Curvature-Dimension I: new model spaces for isoperimetric and concentration inequalities in negative dimension*, Transactions of the American Mathematical Society **369** (2017), no. 5, 3605–3637; arXiv:1409.4109.